CORNERSTONE · IMPLEMENTED TRANSACTION PERSISTENCE

MongoDB transaction document & sync state

One idempotent transaction document is durable at the edge, then replicated to the configured cloud.

BROWSER CLIENT

Local app state

cornerstone-products

cornerstone-customers

cornerstone-cart

cornerstone-sales

Sales queue retries if API is unreachable.

Other app data is not synced by the backend.

Fallback only

Browser storage is not the edge database.

REST API WRITE

LOCAL MONGODB · RASPBERRY PI

transactions collection

DURABLE EDGE RECORD

transactionId	unique idempotency key
payloadHash	SHA-256 content check
createdAt · customer	ISO date · display name
items[]	productId · name · quantity · price
subtotal · tax · total	validated monetary amounts
method	Card | Cash (tender record only)
status	pending | synced
syncedAt	null until cloud copy verified

DIFFERENTIAL SYNC

CONFIGURED AWS CLOUD MONGODB

transactions collection

Same document shape

Upserted by transactionId

payloadHash verified

Set CLOUD_MONGODB_URI to enable.

Unavailable cloud? Edge stays pending.

Cloud copy confirmed

Then local status becomes synced.

VALIDATION & RETRY RULES

Transaction ID, date, customer, line items, integer quantities, supported method and amount totals are checked.

8% tax and item subtotal are checked. Same ID + same payload is safe to retry; same ID + changed data is rejected.

AUTOMATIC RETRY

Every 10 seconds

Manual trigger: POST /api/sync